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BATTERY PACK LIQUID COOLANT GUIDES AND METHOD OF GUIDING LIQUID COOLANT

Abstract

A traction battery pack assembly includes a plurality of battery cell groups disposed along a cell stack axis of a cell stack. Each of the battery cell groups includes at least one battery cell. A plurality of liquid guides are configured to guide a liquid coolant axially between the battery cell groups. The battery cell groups are separated from each other by at least some of the liquid guides within the plurality of liquid guides.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure details exemplary systems that guide liquid coolant within a battery pack and, more particularly, to a system that guides the liquid coolant between cells of a cell stack.

BACKGROUND

[0002] Electrified vehicles differ from conventional motor vehicles because electrified vehicles include a drivetrain having one or more electric machines. The electric machines can drive the electrified vehicles instead of, or in addition to, an internal combustion engine. A traction battery pack assembly can power the electric machines. As part of an immersion thermal management system, liquid coolant can be moved through the traction battery pack to help manage thermal energy within the traction battery pack.

SUMMARY

[0003] In some aspects, the techniques described herein relate to a traction battery pack assembly, including: a plurality of battery cell groups disposed along a cell stack axis of a cell stack, each of the battery cell groups including at least one battery cell; and a plurality of liquid guides configured to guide a liquid coolant axially between the battery cell groups, the plurality of battery cell groups separated from each other by at least some of the liquid guides within the plurality of liquid guides.

[0004] In some aspects, the techniques described herein relate to an assembly, further including a plurality of liquid channels between axially adjacent battery cell groups within the plurality of battery cells.

[0005] In some aspects, the techniques described herein relate to an assembly, further including a plurality of thermal fins, the plurality of battery cell groups separated from each other by at least one of the battery cell groups.

[0006] In some aspects, the techniques described herein relate to an assembly, further including a plurality of liquid channels between axially adjacent battery cell groups within the plurality of battery cells.

[0007] In some aspects, the techniques described herein relate to an assembly, wherein the liquid channels each have a circumferential perimeter established entirely by some of the liquid guides within the plurality of liquid guides and some of the thermal fins within the plurality of thermal fins.

[0008] In some aspects, the techniques described herein relate to an assembly, wherein the thermal fins within the plurality of thermal fins are a metal or metal alloy.

[0009] In some aspects, the techniques described herein relate to an assembly, wherein the liquid guides are each sandwiched between two of the thermal fins within the plurality of thermal fins.

[0010] In some aspects, the techniques described herein relate to an assembly, wherein the liquid guides are compressible.

[0011] In some aspects, the techniques described herein relate to an assembly, wherein the liquid guides are foam.

[0012] In some aspects, the techniques described herein relate to an assembly, wherein the liquid guides are strips that extend from a position adjacent a top side of the cell stack to a position adjacent a bottom side of the cell stack.

[0013] In some aspects, the techniques described herein relate to an assembly, wherein the liquid guides are configured to guide the liquid coolant at a position between axially adjacent battery cells within the plurality of battery cells.

[0014] In some aspects, the techniques described herein relate to an assembly, further including a plurality of stand-offs that maintain a spacing between an enclosure and the plurality of battery cells, the spacing configured to communicate the liquid coolant.

[0015] In some aspects, the techniques described herein relate to an assembly, further including the enclosure, the plurality of stand-offs provided by dimples within the enclosure.

[0016] In some aspects, the techniques described herein relate to an assembly, wherein the plurality

of stand-offs include some of the stand-offs within the plurality of stand-offs vertically beneath the plurality of battery cells and some of the stand-offs within the plurality of stand-offs vertically above the plurality of battery cells.

[0017] In some aspects, the techniques described herein relate to an assembly, wherein the liquid coolant is a dielectric liquid coolant.

[0018] In some aspects, the techniques described herein relate to a method of managing thermal energy levels within a traction battery pack, including: immersing at least a portion of a cell stack within a liquid coolant to manage thermal energy within the cell stack, the cell stack including a plurality of battery cells disposed along a cell stack axis; and guiding the liquid coolant through liquid channels that are axially between groups of the battery cells of the cell stack, the groups spaced axially from each other by a plurality of liquid guides.

[0019] In some aspects, the techniques described herein relate to a method, wherein the plurality of liquid guides are configured to compress in response to expansion of the battery cells along the cell stack axis.

[0020] In some aspects, the techniques described herein relate to a method, further including spacing the cell stack away from an enclosure using a plurality of stand-offs, the spacing providing an area for the liquid coolant to move above the cell stack, beneath the cell stack, or both.

[0021] In some aspects, the techniques described herein relate to a method, wherein the plurality of stand-offs are stamped in the enclosure.

[0022] The embodiments, examples and alternatives of the preceding paragraphs, the claims, or the following description and drawings, including any of their various aspects or respective individual features, may be taken independently or in any combination. Features described in connection with one embodiment are applicable to all embodiments, unless such features are incompatible.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0023] The various features and advantages of the disclosed examples will become apparent to those skilled in the art from the detailed description. The figures that accompany the detailed description can be briefly described as follows:

[0024] FIG. 1 illustrates a side view of an electrified vehicle having a battery pack.

[0025] FIG. 2 illustrates a perspective view of the battery pack of FIG. 1 along with portions of an immersion thermal management system.

[0026] FIG. 3 illustrates an expanded view of the battery pack of FIG. 2.

[0027] FIG. 4 illustrates a close up view of an area in FIG. 3.

[0028] FIG. 5 illustrates a section view taken at line 5-5 in FIG. 2.

DETAILED DESCRIPTION

[0029] An immersion thermal management system can be used to manage thermal energy in a traction battery pack. The immersion thermal management system immerses at least some components of the traction battery pack in a liquid coolant. The immersed components can include a cell stack. This disclosure is directed toward guiding the liquid coolant between cells of the cell stack.

[0030] With reference to FIG. 1, an electrified vehicle **10** includes a traction battery pack **14**, an electric machine **18**, and wheels **22**. The traction battery pack **14** powers an electric machine **18**, which can convert electrical power to mechanical power to drive the wheels **22**. The traction battery pack **14** can be a relatively high-voltage battery.

[0031] The traction battery pack **14** is, in the exemplary embodiment, secured to an underbody **26** of the electrified vehicle **10**. The traction battery pack **14** could be located elsewhere on the electrified vehicle **10** in other examples.

[0032] The electrified vehicle **10** is an all-electric vehicle. In other examples, the electrified vehicle **10** is a hybrid electric vehicle, which selectively drives wheels using torque provided by an internal combustion engine instead of, or in addition to, an electric machine. Generally, the electrified vehicle **10** could be any type of vehicle having a traction battery pack.

[0033] Although the different examples have the specific components shown in the illustrations, embodiments of this disclosure are not limited to those particular combinations. It is possible to use some of the components or features from one of the examples in combination with features or components from another one of the examples. In addition, the various figures accompanying this disclosure are not necessarily to scale, and some features may be exaggerated or minimized to show certain details of a particular component or arrangement.

[0034] FIG. **2** illustrates additional detail of the example battery pack **14**. In this example, the battery pack **14** includes an enclosure assembly **30**. The enclosure assembly **30** includes a cover **34** and a tray **38**. The cover **34**, in this example, is vertically above the tray **38**. In other examples, however, the cover **34** could be arranged below, or to a side of the tray **38**. Various terms such as “above,” “below,” “top,” and “bottom” are used relative to the arrangement of the components of the battery pack **14** in the various drawings and should not otherwise be deemed limiting. These terms are with reference to the general orientation of the battery pack **14** when installed within the vehicle **10** of FIG. **1**,

[0035] The cover **34** is welded to the tray **38** in one example of this disclosure. While welding is mentioned, the cover **34** and tray **38** could be connected using other fluid-tight connection techniques, such as adhesive. Further, while an exemplary enclosure assembly **30** is shown in the drawings, the enclosure assembly **30** may vary in size, shape, and configuration within the scope of this disclosure.

[0036] In this disclosure, a cell stack **42** is arranged within the enclosure assembly **30**. The cell stack **42** includes a plurality of individual battery cells **46** disposed along a cell stack axis **A** between a pair of endplates **50**. The cell stack **42** could include any number of battery cells **46**. The battery pack **14** could employ any number of cell stacks **42** within the enclosure assembly **30**. Thus, this disclosure is not limited to the exact configuration shown in FIG. **2**. Further, while the battery cells **46** of FIG. **2** are positioned side-by-side relative to one another, other configurations are also contemplated within the scope of this disclosure, including but not limited to embodiments in which the battery cells **46** are stacked on top of one another, for example.

[0037] In an embodiment, the battery cells **46** are lithium-ion pouch cells. However, battery cells having other geometries (cylindrical, etc.), other chemistries (nickel-metal hydride, lead-acid, etc.), or both could alternatively be utilized within the scope of this disclosure.

[0038] The cell stack **42** is arranged in an interior of the enclosure assembly **30** within the tray **38** and beneath the cover **34**. A thermal management system is used to manage thermal energy levels within the battery pack **14**. The example thermal management system is configured to route non-conductive (i.e., dielectric) liquid coolant **C** over areas of the cell stack **42** to manage thermal energy within the cell stack **42** by, for example, using the coolant **C** to take on heat from the cell stack **42**. the thermal management system is an immersion thermal management system at least because portions of the battery pack **14**, here at least the battery cells **46** of the cell stack **42** are immersed in the coolant **C**.

[0039] In this example, the coolant **C** generally flows from an inlet **52**, which is formed in the cover **34**, to an outlet **54**, which is formed in the tray **38** at an opposite end of the enclosure assembly **30** from the inlet **52**. The cell stack **42** is spaced from portions of the tray **38** with at least one stand-off **58**. The cell stack **42** is spaced from portions of the cover **34** with at least one stand-off **62**. The spacing provided by the stand-offs **58** and **62** provides areas for the coolant **C** to move to the outlet **54** above the cell stack **42** and beneath the cell stack **42**.

[0040] In this example, the cover **34** and the tray **38** are metal or metal alloys. The stand-offs **58** and **62** can be, for example, dimples that are stamped in the cover **34** and the tray **38**, respectively.

In another example, the stand-offs **58** and **62** are separate pieces, such as foam strips, that are adhered to the cover **34** and the tray **38**.

[0041] With reference now to FIGS. **3-5**, the cell stack **42** has a top side **66**, opposing outboard sides **70**, and a bottom side **74**. Within the interior of the enclosure assembly **30**, some of the coolant **C** moves from the cover **34** through the top side **66** of the cell stack **42** and in between the cells **46**. Thermal energy can transfer between the coolant and the cells **46** as the coolant **C** moves between the cells **46**. As can be appreciated, increasing a surface area of the cells **46** contacting the coolant **C** can facilitate thermal energy transfer.

[0042] To maintain spacing for the coolant to move through the cell stack **42**, the cell stack **42** incorporates a plurality of liquid guides **78** that keeps groups **82** of the cells **46** spaced from each other to establish liquid channels **86** between the groups **82** of the cells **46**. The coolant **C** can communicate vertically downward through the cells stack **42** within the liquid channels **86** that provides clearance between the cells **46** for the coolant **C** to move between the cells **46**.

[0043] In this example, the groups **82** each include two of the cells **46**. In other examples, other numbers of cells **46** could be included in the groups **82**-the groups **82** could even each include one cell **46**. Each of the groups **82** is spaced from the axially adjacent groups **82** by five of the liquid guides **78** to establish four liquid channels **86** between each of the axially adjacent groups **82** within the cells stack **42**. Additional liquid guides **78** are sandwiched between each endplate **50** and groups **82** of the cells **46** to establish liquid channels **86** at the axial ends of the cell stack **42**.

[0044] The liquid guides **78** are strips that extend from a position adjacent the top side **66** to the of the cell stack **42** to the bottom side **74** of the cell stack **42**.

[0045] The liquid guides **78** are compressible in this example. The liquid guides **78** can be foam, aerogel, or both. During operation, should cells **46** expand along the cell stack axis **A**, the liquid guides **78** can, in response, be compressed to accommodate expansion of the cells **46**.

[0046] The cell stack **42** includes a plurality of thermal fins **90**. In this example, each of the cells **46** interfaces directly with one of the thermal fins **90**. The liquid guides **78** are each sandwiched between two of the thermal fins **90**. The thermal fins **90** can facilitate pressure uniformity when compressing the liquid guides **78** due to expansion of one or more of the cells **46** along the axis **A**. The thermal fins **90** can facilitate thermal transfer to the liquid within the liquid channels **86**. The thermal fins **90** can provide a physical barrier to block particulates vented out from the cells **46** during thermal propagation.

[0047] As the thermal fins **90** sandwich the liquid guides **78**, the liquid channels **86** of the exemplary embodiment each have a circumferential perimeter established entirely by two of the liquid guides **78** and two of the thermal fins **90**.

[0048] Features of disclosed examples include a system that provides space for flow between cells, and that guides flow moving between cells.

[0049] The preceding description is exemplary rather than limiting in nature. Variations and modifications to the disclosed examples may become apparent to those skilled in the art that do not necessarily depart from the essence of this disclosure. Thus, the scope of protection given to this disclosure can only be determined by studying the following claims.

Claims

1. A traction battery pack assembly, comprising: a plurality of battery cell groups disposed along a cell stack axis of a cell stack, each of the battery cell groups including at least one battery cell; and a plurality of liquid guides configured to guide a liquid coolant axially between the battery cell groups, the plurality of battery cell groups separated from each other by at least some of the liquid guides within the plurality of liquid guides.

2. The assembly of claim 1, further comprising a plurality of liquid channels between axially adjacent battery cell groups within the plurality of battery cells.

- 3.** The assembly of claim 1, further comprising a plurality of thermal fins, the plurality of battery cell groups separated from each other by at least one of the battery cell groups.
 - 4.** The assembly of claim 3, further comprising a plurality of liquid channels between axially adjacent battery cell groups within the plurality of battery cells.
 - 5.** The assembly of claim 4, wherein the liquid channels each have a circumferential perimeter established entirely by some of the liquid guides within the plurality of liquid guides and some of the thermal fins within the plurality of thermal fins.
 - 6.** The assembly of claim 3, wherein the thermal fins within the plurality of thermal fins are a metal or metal alloy.
 - 7.** The assembly of claim 3, wherein the liquid guides are each sandwiched between two of the thermal fins within the plurality of thermal fins.
 - 8.** The assembly of claim 1, wherein the liquid guides are compressible.
 - 9.** The assembly of claim 1, wherein the liquid guides are foam.
 - 10.** The assembly of claim 1, wherein the liquid guides are strips that extend from a position adjacent a top side of the cell stack to a position adjacent a bottom side of the cell stack.
 - 11.** The assembly of claim 1, wherein the liquid guides are configured to guide the liquid coolant at a position between axially adjacent battery cells within the plurality of battery cells.
 - 12.** The assembly of claim 1, further comprising a plurality of stand-offs that maintain a spacing between an enclosure and the plurality of battery cells, the spacing configured to communicate the liquid coolant.
 - 13.** The assembly of claim 12, further comprising the enclosure, the plurality of stand-offs provided by dimples within the enclosure.
 - 14.** The assembly of claim 13, wherein the plurality of stand-offs include some of the stand-offs within the plurality of stand-offs vertically beneath the plurality of battery cells and some of the stand-offs within the plurality of stand-offs vertically above the plurality of battery cells.
 - 15.** The assembly of claim 1, wherein the liquid coolant is a dielectric liquid coolant.
 - 16.** A method of managing thermal energy levels within a traction battery pack, comprising: immersing at least a portion of a cell stack within a liquid coolant to manage thermal energy within the cell stack, the cell stack including a plurality of battery cells disposed along a cell stack axis; and guiding the liquid coolant through liquid channels that are axially between groups of the battery cells of the cell stack, the groups spaced axially from each other by a plurality of liquid guides.
 - 17.** The method of claim 16, wherein the plurality of liquid guides are configured to compress in response to expansion of the battery cells along the cell stack axis.
 - 18.** The method of claim 16, further comprising spacing the cell stack away from an enclosure using a plurality of stand-offs, the spacing providing an area for the liquid coolant to move above the cell stack, beneath the cell stack, or both.
 - 19.** The method of claim 18, wherein the plurality of stand-offs are stamped in the enclosure.
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